

Chemistry Ch 3.4: Relative Atomic Mass

Chapter 3, Atomic Structure · kitab ke page 27 se 44

Paimana: Carbon-12

Atom itna chhota hai ke us ka mass gram mein likhna na mumkin hai. Is liye hum use kisi aur se **mila kar** naapte hain, aur wo paimana hai **C-12**, jo **1961** se standard hai.

1 amu = ek C-12 atom ke mass ka 1/12 hissa.

Relative atomic mass yani kisi atom ka mass, C-12 ke 1/12 hisse ke muqable mein.

Kitab ki table ke numbers

H	N	O	Na	Al	S	Cl	Fe
1.008	14.0067	15.9994	22.9898	26.9815	32.06	35.453	55.847

Ye number poore (whole) kyun nahin hain

Chlorine ka 35.453 hai, 35 nahin. Kyun?

Kyunke qudrat mein chlorine do shaklon mein milta hai: **Cl-35** aur **Cl-37**. Jo number table mein hai wo dono ka **auzan ke saath aosat (weighted average)** hai.

Yani ye number kisi ek atom ka nahin, poori qudrati mixture ka hai. Yehi agle safhay, isotopes, ka raasta hai.

Quiz

1. Define **relative atomic mass** and **1 amu**. Which atom is the standard, and since which year?
2. The relative atomic mass of chlorine is 35.453, not a whole number. Explain why.